Software Requirements Specification

for

Financial Scam Detection, Investigation, and Recovery Assistance System

Version 1.0 approved

Team Details

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Project ID: 35

Course: UE24CS341A – Software Engineering

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Revision History

Name	Date	Reason for Changes	Version
Team (Project ID 35)	September 2026	Initial draft of SRS	1.0

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document describes the functional and non-functional requirements for Version 1.0 of the Financial Scam Detection, Investigation, and Recovery Assistance System (FSDIRAS). The scope of this document covers the complete system, comprising four integrated subsystems: the Scam Detection Engine, the Investigation and Evidence Management Module, the Recovery and Reporting Assistance Module, and the Education, Prevention and Administration Module. This document is intended to serve as the primary reference for the design, development, and testing phases of the project undertaken as part of the course UE24CS341A – Software Engineering.

1.2 Intended Audience and Reading Suggestions

This document is intended for the following readers:

- Development Team – to understand the functional scope and design constraints before implementation.
- Course Instructors / Evaluators – to assess the completeness and correctness of the requirements against the delivered system.
- Testers – to derive test cases and validate that the implemented system meets specified requirements.
- End Users (Victims / General Public, Investigators, Bank Recovery Officers, Administrators) – to understand the capabilities the system will provide to them.

The remainder of this document is organized as follows: Section 2 provides an overall description of the product; Section 3 specifies external interface requirements; Section 4 presents analysis models; Section 5 details system features and their associated functional requirements; Section 6 specifies non-functional requirements; Section 7 lists other requirements; and the Appendices provide a glossary, field layouts, and a requirements traceability matrix.

1.3 Product Scope

Financial scams such as phishing messages, fraudulent UPI payment requests, fake loan applications, and investment fraud have grown substantially, causing significant monetary loss and emotional distress to victims, while overwhelming law enforcement and bank fraud desks with unstructured complaints. FSDIRAS is a web-based platform designed to (a) automatically detect and flag suspicious messages, URLs, and transaction patterns using AI/ML techniques, (b) allow victims to report scams and submit supporting evidence through a structured workflow, (c) enable investigators to manage cases with a verifiable chain of custody for evidence, (d) assist in tracking fund-recovery efforts and generate standardized reports for submission to banks, the National Cyber Crime Reporting Portal (NCRP), or the RBI Ombudsman, and (e) educate the public on common scam patterns to reduce future incidents. The overall objective is to reduce the average time from scam occurrence to reporting and recovery, improve the quality and traceability of evidence, and increase public awareness of financial fraud.

1.4 References

- IEEE Recommended Practice for Software Requirements Specifications (IEEE Std 830-1998) — SRS template structure (Karl Wiegers).
- National Cyber Crime Reporting Portal, https://cybercrime.gov.in

- Reserve Bank of India — Master Directions on Fraud Risk Management and Ombudsman Scheme, https://www.rbi.org.in
- Information Technology Act, 2000 (India) and subsequent amendments.
- Project Plan Document and Test Plan Document, UE24CS341A – Software Engineering, Project ID 35.

2. Overall Description

2.1 Product Perspective

FSDIRAS is a new, self-contained web application and is not a replacement for any existing government system such as the NCRP; rather, it is designed to complement such systems by providing a faster, AI-assisted front end for detection and case management, with manual export/handoff of finalized reports to official channels in this phase of the project. The system consists of four major subsystems that interact with a shared database and a common user-management layer:

- Scam Detection Engine (AI/ML) — analyzes submitted text, URLs, and transaction metadata to produce a risk score.
- Investigation & Evidence Management Module — manages case lifecycle and chain-of-custody for evidence.
- Recovery & Reporting Assistance Module — tracks fund-recovery status and generates standardized reports.
- Education, Prevention & Administration Module — hosts awareness content and provides system administration.

2.2 Product Functions

- User registration and role-based login (Victim, Investigator, Recovery Officer, Administrator).
- Submission of scam reports along with evidence such as screenshots, transaction IDs, and chat logs.
- AI/ML-based risk scoring of reported messages, URLs, and transaction patterns.
- Automatic and manual case creation and assignment to investigators.
- Evidence upload with an immutable, timestamped chain-of-custody log.
- Investigation status tracking, case notes, and escalation workflow.
- Recovery-assistance request generation and status tracking with banks/payment gateways.
- Automated, downloadable case and recovery report generation (PDF).
- Awareness content, articles, and short quizzes for scam-prevention education.
- Administration dashboard for user, role, content, and blacklist/whitelist management.
- Email/SMS notifications to users on status changes.

2.3 User Classes and Characteristics

User Class	Characteristics
Victim / Reporting User	General public with basic digital literacy; primary use is submitting a scam report and tracking its status; infrequent, occasional use.
Investigator / Case Officer	Trained personnel (bank fraud desk / cyber-cell liaison) responsible for reviewing evidence and updating case status; frequent, daily use; requires elevated access.
Recovery / Bank Liaison Officer	Personnel coordinating with banks/payment gateways to track recovery of funds; moderate frequency of use.
System Administrator	Manages users, roles, awareness content, and detection blacklists/whitelists; occasional but high-privilege use.
Guest	Unauthenticated visitor who can only view public awareness/education content.

2.4 Operating Environment

FSDIRAS is delivered as a responsive web application accessible through modern browsers (Chrome, Firefox, Edge) on desktop and mobile devices. The backend is deployed on a Linux-based server, using a relational database (PostgreSQL/MySQL) for structured data and encrypted object storage for evidence files. The AI/ML scam-detection service is exposed as an internal REST API (Python/FastAPI or Flask) consumed by the main application server. The system is designed to be containerized (Docker) to allow deployment on cloud infrastructure such as AWS or Azure.

2.5 Design and Implementation Constraints

- The system must comply with applicable provisions of the Information Technology Act, 2000, and relevant RBI guidelines on data handling and fraud reporting.
- Evidence records must be immutable once submitted, to preserve chain-of-custody integrity for potential legal use.
- The system shall use open-source frameworks and libraries wherever feasible, consistent with the academic scope of the project.
- The ML detection component shall be a modular service so the underlying model can be updated or retrained independently of the main application.
- All communication between the client and server shall be secured using HTTPS/TLS.

2.6 Assumptions and Dependencies

- Users are assumed to have basic internet access and the ability to operate a web browser.
- The system depends on a third-party email/SMS gateway API (e.g., SendGrid/Twilio or equivalent) for notifications; unavailability of this service will degrade the notification feature but not core functionality.
- The ML detection model is trained on a labeled dataset assembled from publicly available scam/phishing datasets and synthetic data generated for the project; production-grade accuracy is not guaranteed and the score is advisory.
- Direct system-to-system integration with bank cores or the NCRP is out of scope for this phase; recovery/report handoff is assumed to be a manual, human-mediated step.

3. External Interface Requirements

3.1 User Interfaces

The system provides distinct dashboards for each user role (Victim, Investigator, Recovery Officer, Administrator). A guided, multi-step wizard is provided for scam report submission, with inline validation of mandatory fields. All screens share a common header with navigation, a help/FAQ link, and a logout control. Error messages are displayed inline, close to the relevant field, using plain, non-technical language. Detailed screen mock-ups are to be maintained in a separate UI design document.

3.2 Software Interfaces

Component	Description
Database	PostgreSQL/MySQL — stores user accounts, case records, evidence metadata, recovery status, and awareness content.
ML Detection Service	Internal REST API (JSON over HTTPS) serving risk-score predictions from the trained model.
Evidence Storage	Encrypted file storage (e.g., AWS S3 or equivalent) for uploaded screenshots, PDFs, and chat-log exports.
Email/SMS Gateway	Third-party API used to dispatch status-change notifications to users.
Report Generator	Internal service that compiles case data into a formatted PDF report.

3.3 Communications Interfaces

All client-server communication uses HTTPS with TLS 1.2 or higher. Internal services communicate over REST APIs using JSON payloads. Email notifications use SMTP via the configured gateway; SMS notifications use the gateway's HTTP API. File uploads are transmitted as multipart/form-data and are size-limited and virus-scanned before storage.

4. Analysis Models

The primary actors identified for the system are: Victim/Reporting User, Investigator, Recovery Officer, Administrator, and Guest. Key use cases include: Register/Login, Submit Scam Report, Analyze Risk Score, Auto/Manual Case Assignment, Upload and Track Evidence, Update Investigation Status, Generate Recovery Request, Track Recovery Status, Generate Case Report, View Awareness Content, Take Awareness Quiz, and Manage Users/Content/Blacklist. A detailed use-case diagram and supporting entity-relationship diagram for the data model (User, Report, Case, Evidence, RecoveryRequest, AwarenessContent entities and their relationships) are maintained as separate design artifacts and will be included in the project's design documentation.

5. System Features

5.1 Scam Detection & AI/ML Module

5.1.1 Description and Priority

This feature uses a trained machine-learning classification model to analyze submitted text messages, URLs, and transaction metadata, assigning a risk score that indicates the likelihood that the item is a scam. It also cross-checks submissions against a maintained blacklist of known scam URLs, phone numbers, and UPI IDs. Priority: High.

5.1.2 Stimulus/Response Sequences

The user submits a message, URL, or transaction detail for analysis. The system extracts relevant features, checks them against the blacklist, and passes them to the ML model. The model returns a risk score, which the system displays to the user along with a brief explanation. If the score exceeds the high-risk threshold, the system automatically creates an investigation case.

5.1.3 Functional Requirements

REQ-1: The system shall accept free-text messages, URLs, or transaction metadata as input for risk analysis.

REQ-2: The system shall compute a risk score on a 0–100 scale using the trained ML model within 5 seconds for 95% of requests.

REQ-3: The system shall categorize each analyzed item into Low, Medium, or High risk bands based on configurable thresholds.

REQ-4: The system shall maintain and check submissions against a blacklist of known scam URLs, phone numbers, and UPI IDs prior to invoking the ML model.

REQ-5: The system shall automatically create an investigation case when a submission is classified as High risk.

REQ-6: The system shall log every detection request and its result for audit purposes and future model retraining.

REQ-7: The system shall allow an administrator to trigger periodic retraining or updating of the ML model using newly labeled data.

5.2 Investigation & Evidence Management Module

5.2.1 Description and Priority

This feature enables investigators to manage the lifecycle of a case created from a scam report, including reviewing and organizing evidence with a verifiable chain of custody, recording investigative notes, and updating case status. Priority: High.

5.2.2 Stimulus/Response Sequences

A case is created (automatically by the detection module or manually by a victim's report). The case is assigned to an investigator, who reviews and uploads evidence, records notes, and updates the case status until it is resolved, escalated, or closed.

5.2.3 Functional Requirements

REQ-8: The system shall auto-generate a unique Case ID for every scam report received.

REQ-9: The system shall support assignment of a case to an investigator, either manually by an administrator or automatically via round-robin allocation.

REQ-10: The system shall allow investigators and victims to upload evidence files (screenshots, PDFs, chat-log exports) up to a configurable size limit.

REQ-11: The system shall maintain an immutable, timestamped audit log recording the user, action, and time for every evidence item, to preserve chain of custody.

REQ-12: The system shall support the case status values: New, Under Investigation, Escalated, Resolved, and Closed.

REQ-13: The system shall prevent deletion of submitted evidence; evidence may only be archived with a recorded reason, never removed.

REQ-14: The system shall allow investigators to add timestamped, author-attributed notes to a case.

5.3 Recovery & Reporting Assistance Module

5.3.1 Description and Priority

This feature assists victims and investigators in tracking the progress of fund recovery with banks or payment gateways, and generates standardized, downloadable reports summarizing a case for submission to banks, the police, or the RBI Ombudsman. Priority: High.

5.3.2 Stimulus/Response Sequences

Once a case is deemed ready for recovery action, the investigator marks it accordingly. The system auto-populates a recovery-assistance request from case data. The recovery status is updated over time, and the victim is notified of changes. On closure, a final report is generated.

5.3.3 Functional Requirements

REQ-15: The system shall auto-populate a recovery-assistance request template using data already captured in the case record.

REQ-16: The system shall allow authorized users to record and update recovery status as Requested, In Progress, Partially Recovered, Fully Recovered, or Rejected.

REQ-17: The system shall generate a downloadable PDF report of the case summary, including timeline, evidence list, and risk score.

REQ-18: The system shall send a notification to the victim whenever the recovery status of their case changes.

REQ-19: The system shall track and display the amount reported as lost versus the amount recovered for each case.

5.4 Education, Prevention & Administration Module

5.4.1 Description and Priority

This feature provides categorized awareness content and short quizzes to educate the public on common financial scams, along with administrative functions for managing users, roles, content, and detection blacklists/whitelists. Priority: Medium.

5.4.2 Stimulus/Response Sequences

A guest or registered user browses the awareness section, reads articles, and may take a short quiz. Separately, an administrator logs in to the admin panel to manage users, content, or blacklist/whitelist entries and to view system-wide analytics.

5.4.3 Functional Requirements

REQ-20: The system shall display categorized awareness articles covering common scam types such as phishing, UPI fraud, and fake loan applications.

REQ-21: The system shall provide short, scored quizzes that allow users to test their awareness of scam-recognition techniques.

REQ-22: The system shall allow an administrator to create, edit, and remove awareness content.

REQ-23: The system shall allow an administrator to manage user accounts and roles, including activation, deactivation, and role assignment.

REQ-24: The system shall provide an analytics dashboard to administrators showing total reports, resolved cases, recovery rate, and trending scam categories.

REQ-25: The system shall allow an administrator to add, edit, or remove entries in the blacklist/whitelist used by the detection module.

6. Other Nonfunctional Requirements

6.1 Performance Requirements

- The scam-detection risk-scoring response shall be returned within 5 seconds for at least 95% of requests.
- The system shall support at least 500 concurrent users without degradation of core functionality.
- Web pages shall load within 3 seconds under normal network conditions.

6.2 Safety Requirements

- Evidence data shall be backed up on a daily basis to prevent data loss.
- The ML-generated risk score shall always be presented as advisory guidance rather than a conclusive determination, to prevent wrongful accusation based solely on an automated score.
- Case status changes that could affect a victim's legal recourse (e.g., closure) shall require explicit investigator confirmation.

6.3 Security Requirements

- All data in transit shall be encrypted using TLS 1.2 or higher.
- Passwords shall be stored using a strong one-way hashing algorithm (e.g., bcrypt).
- Access to case and evidence data shall be governed by role-based access control, restricted to the assigned investigator, relevant recovery officer, and administrators.
- Evidence files shall be encrypted at rest.
- All security-relevant actions (login, evidence access, status changes) shall be recorded in an audit log.

6.4 Software Quality Attributes

- Usability: The scam-reporting workflow shall be completable by a first-time user in under 5 minutes without external assistance.
- Reliability: The system shall target 99% uptime during the evaluation period.
- Maintainability: The system shall follow a modular design so that the detection, investigation, recovery, and education subsystems can be updated independently.
- Scalability: The ML detection service shall be independently scalable (horizontally) from the main application server.
- Portability: The application shall be containerized to allow deployment across different cloud environments.

6.5 Business Rules

- Only users with the Investigator role may update the status of a case assigned to them.
- Evidence, once submitted, cannot be edited or deleted — only appended to or archived with a documented reason.
- The ML-generated risk score is advisory; a final scam determination requires human review by an investigator.

- A victim's personally identifiable information (PII) is visible only to the assigned investigator, relevant recovery officer, and system administrators.

7. Other Requirements

- Database records related to cases and evidence shall be retained for a minimum of 5 years, consistent with typical financial record-keeping norms, subject to applicable regulation.
- Multilingual support (English plus at least one regional language) is a planned enhancement for a future phase and is out of scope for Version 1.0.
- The system shall display a legal disclaimer clarifying that use of the platform does not guarantee recovery of lost funds and does not replace filing an official complaint with the police or the National Cyber Crime Reporting Portal.

Appendix A: Glossary

Term	Definition
Scam	A deceptive scheme intended to defraud a victim of money or sensitive information.
Phishing	A social-engineering technique that uses fraudulent communication (message, email, or website) to trick a user into revealing sensitive information.
UPI	Unified Payments Interface — a real-time payment system widely used in India.
Chain of Custody	A chronological, timestamped record documenting the handling of evidence, used to establish its integrity and admissibility.
Risk Score	A numeric value (0–100) produced by the ML model indicating the likelihood that a submitted item is fraudulent.
NCRP	National Cyber Crime Reporting Portal — the Government of India's official portal for reporting cybercrime.
RBI Ombudsman	A grievance-redressal mechanism provided by the Reserve Bank of India for resolving customer complaints against banks.
Case	A structured investigation record created from one or more related scam reports.
Evidence	Any file or data item (screenshot, transaction record, message log) submitted in support of a scam report.
PII	Personally Identifiable Information — data that can be used to identify a specific individual.
ML Model	The machine-learning model trained to classify submissions by risk level.
Blacklist	A maintained list of known-fraudulent URLs, phone numbers, or UPI IDs used to flag submissions.
Recovery Assistance	The process of tracking and facilitating the return of funds lost to a scam via banks or payment gateways.

Appendix B: Field Layouts

The following tables define the field layout for the primary Scam Report entity, and the fields to be included in the generated Case and Recovery reports.

Scam Report — Field Layout

Field	Length	Data Type	Description	Is Mandatory
Report ID	12	Alphanumeric	System-generated unique identifier for the report	Y
Reporter Name	60	String	Full name of the person submitting the report	Y
Reporter Contact	15	Alphanumeric	Phone number or email of the reporter	Y
Report Date	8	Date	Date the report was submitted	Y
Scam Category	30	Alphanumeric	Type of scam (phishing, UPI fraud, loan app, investment, other)	Y
Description	500	String	Free-text description of the incident	Y
Amount Involved	12	Numeric	Monetary amount reportedly lost	N
Evidence Attached	1	Boolean	Indicates whether supporting evidence was uploaded	N
Risk Score	3	Numeric	ML-computed risk score (0–100)	N
Status	25	Alphanumeric	Current case status	Y

Sample Report Requirements

Case Report	Recovery Report
Case ID	Case ID
Reporter Name	Reporter Name / Account Number
Scam Category	Bank Name
Report Date	IFSC Code
Risk Score	Amount Reported
Investigator Assigned	Amount Recovered
Evidence List	Recovery Status
Case Status	Remarks

Appendix C: Requirement Traceability Matrix

Sl. No	Requirement ID	Brief Description	Related Feature	Test Case ID
1	REQ-1	Accept text/URL/transaction input for analysis	Scam Detection & AI/ML	TBD
2	REQ-2	Compute risk score within 5 seconds	Scam Detection & AI/ML	TBD
3	REQ-3	Categorize into Low/Medium/High risk	Scam Detection & AI/ML	TBD
4	REQ-4	Maintain and check blacklist	Scam Detection & AI/ML	TBD
5	REQ-5	Auto-create case on high risk	Scam Detection & AI/ML	TBD
6	REQ-6	Log detection requests for audit	Scam Detection & AI/ML	TBD
7	REQ-7	Admin-triggered model retraining	Scam Detection & AI/ML	TBD
8	REQ-8	Auto-generate unique Case ID	Investigation & Evidence Mgmt	TBD
9	REQ-9	Assign case to investigator	Investigation & Evidence Mgmt	TBD
10	REQ-10	Upload evidence files	Investigation & Evidence Mgmt	TBD
11	REQ-11	Immutable chain-of-custody log	Investigation & Evidence Mgmt	TBD
12	REQ-12	Case status states	Investigation & Evidence Mgmt	TBD
13	REQ-13	Prevent evidence deletion	Investigation & Evidence Mgmt	TBD
14	REQ-14	Timestamped case notes	Investigation & Evidence Mgmt	TBD
15	REQ-15	Auto-populate recovery request	Recovery & Reporting Assistance	TBD
16	REQ-16	Update recovery status	Recovery & Reporting Assistance	TBD
17	REQ-17	Generate downloadable PDF report	Recovery & Reporting Assistance	TBD
18	REQ-18	Notify victim on status change	Recovery & Reporting Assistance	TBD
19	REQ-19	Track reported vs recovered amount	Recovery & Reporting Assistance	TBD
20	REQ-20	Display awareness articles	Education, Prevention & Admin	TBD
21	REQ-21	Provide scored quizzes	Education, Prevention & Admin	TBD
22	REQ-22	Admin manages content	Education, Prevention & Admin	TBD
23	REQ-23	Admin manages users/roles	Education, Prevention & Admin	TBD

24	REQ-24	Analytics dashboard	Education, Prevention & Admin	TBD
25	REQ-25	Manage blacklist/whitelist	Education, Prevention & Admin	TBD